

Accurate Semantic Segmentation of Aerial Imagery Using Attention Res U-Net Architecture

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Abstract— Semantic segmentation of photos holds importance in various fields such, as security, emergency response, agriculture, urban planning and surveillance. Researchers have shown increased interest in automating the analysis of photographs without intervention due to its critical role across multiple disciplines. A CNN variant called U-Net has been successful in segmenting photos beyond the domain. However U-Nets limited number of layers hampers its ability to extract information from photos accurately and can lead to flawed boundaries for objects with complex features. In this study we introduce an architecture called Attention Res U-Net that addresses challenges in semantic segmentation such as low accuracy, for high resolution images and incorrect object boundary recognition. Our approach focuses on refining object boundaries. This approach successfully balances the extraction of attributes with the preservation of space thereby significantly improving aerial image analysis.

Keywords— Attention U-Net architecture mechanisms, U-Net architecture, Semantic segmentation, Aerial imagery.

I. INTRODUCTION

In deep learning algorithm, every pixel in an image is associated with semantic segmentation which is necessary for recognizing and classifying groups of pixels based on their characteristics. Monitoring of environment, disaster management and planning in urban areas are applications of semantic segmentation [5]. Accurately identifying and delineating objects like roads buildings and vegetation is vital for extracting information, from these images [18]. The advent of learning has revolutionized computer vision with neural networks emerging as highly effective tools in

many ways of picture segmentation jobs [5]. The U-Net framework first attracted attention when it was suggested for natural photo segmentation [4], as it could precisely define the borders of objects. To overcome these complexities a tailored approach becomes necessary. Attention processes have shown to be effective, in enhancing segmentation accuracy and improving feature extraction, in computer vision tasks [11] [3].

In light of this we propose an Attention Res U-Net structure particularly tailored for semantic segmentation. Our contributions are as follows;

- Including attention and residual mechanisms in the design, so the model can identify and prioritize different parts of the image.
- Enhancing the performance of segmentation, on aerial imaging datasets.
- Evaluating our proposed framework using a dataset and comparing it to existing methods.

Attention Res U-Net Architecture:

The designing of U-Net model is developed as well as equipped for segmentation of pictures biomedical sector, and has gained popularity because of its ability to accurately identify object boundaries. The U-Nets encoder decoder structure with connections allows for feature identification and reconstruction. In this study we expanded on the Res U-Nets attention mechanisms to enhance its segmentation performance for imagery. By incorporating attention processes we can extract features while maintaining information and focusing on areas. Figure 1 [20] illustrates the design of our Attention Res U model. Our

implementation of the Attention Res U framework utilizes attention features to evaluate both local and global contexts. This enables us to effectively

divide imagery addressing challenges such, as complex scenes and varying illumination conditions.

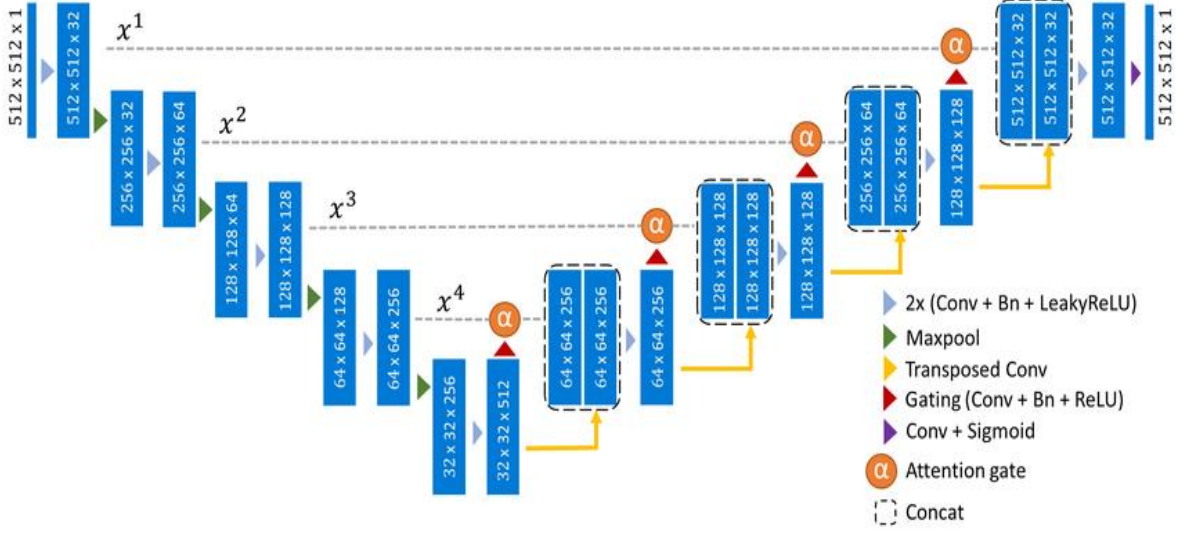


Fig. 1. Design of Attention Res U-Net.

II. RELATED WORK

In the field of perpendicular vision there has been advancement, semantic segmentation credits to the groundbreaking impact on mechanisms of deep learning. However contemporary studies have employed learning techniques to achieve precise results. Ronneberger et al represented the first U-Net design [4]. It quickly gained traction for segmentation tasks, and researchers have since refined it even more to capture global correlations more successfully. Ruigang Niu et al. developed the attention network (HMANet) [9] which takes into account category, space and channel perspectives. Zhongbin Su et al. proposed a neural network (DCNN) that combines DenseNet, U-Net and dilated convolutions to enhance performance. Irwansyah et al. [14] presented a learning based segmentation model specifically designed for identifying buildings in areas. Researchers developed an architecture called Seg-U-net, which combines the Segnet and Unet designs to extract structures from high resolution aerial photographs. This improvement led to the creation of MultiRes U-Net.

III. METHODOLOGY

A. Preparation of Data

We have considered using the Semantic Segmentation dataset available, on Kaggle to test the suggested model. This dataset consists of pictures gathered from accessible sources. To

access the segmentation of photography you can visit this link: <https://rb.gy/5h1uuuj>. The photos captured by satellites named MBRSC, contains level semantic segmentation labels for six different classifications which includes the dataset that holds the aerial pictures of Dubai. It comprises 72 photos divided into six tiles. Locations and conditions. The accurate annotations provided in this dataset played a role, in our decision to select it. We set the resolution of every image to the same level when developing and assessing our classification model in order to guarantee consistency and make integrating into the algorithm easier.

B. Network Architecture

U framework, originally developed for segmenting images the U-Net architecture has gained recognition in domains, including aerial photo analysis. It serves as a component, for our objective due to its skip connections that help maintain spatial context and encoder decoder structure. To further enhance segmentation accuracy.

C. Training Methodology

A significant part of our strategy involved modifying the entropy loss. This modification took into account class weights, which played a role, in enabling learning across all classes. By assigning priority, to classes the model was able to learn more effectively. For testing and validation purposes twenty percent of the data was used, while eighty percent was dedicated to training.

D. Evaluation Criteria

To fully examine our model's efficiency, we made extensive use of a range of assessment metrics. Pixel precision and first metric measures the percentage of recognition of picture element in the images. This provides an evaluation of our models accuracy on a level. This ratio shows how decently each class is carried out by our algorithm. To provide an assessment of segmentation accuracy we calculate the IoU (mIoU) by summing up IoU scores for all classes.

D. Setup for the Experiment

During the testing phase we took consideration of the training collections, software and hardware. To train our Attention architecture successfully we utilized a robust computing system equipped with a 7.9 GB GPU, 8 GigaBytes of RAM and an Intel i7 processor. This smart combination allowed us to

make use of the processing capabilities of the GPUs for model training purposes. The construction of our model involved utilizing software frameworks.

E. Visualization and Data Analysis

We effectively communicated the outcomes of segmentation using visualization techniques, like segmentation maps well as validation and training loss graphs. We were able to confirm that our **Attention Res U-Net architecture is highly effective in achieving segmentation in aerial photographs**, which was a key focus of our study. You can see our suggested model represented in Fig 2. Difference between training & validation loss for the three architectures shown in Fig 4, while Figure 3 displays training & validation IoU, for each architecture.

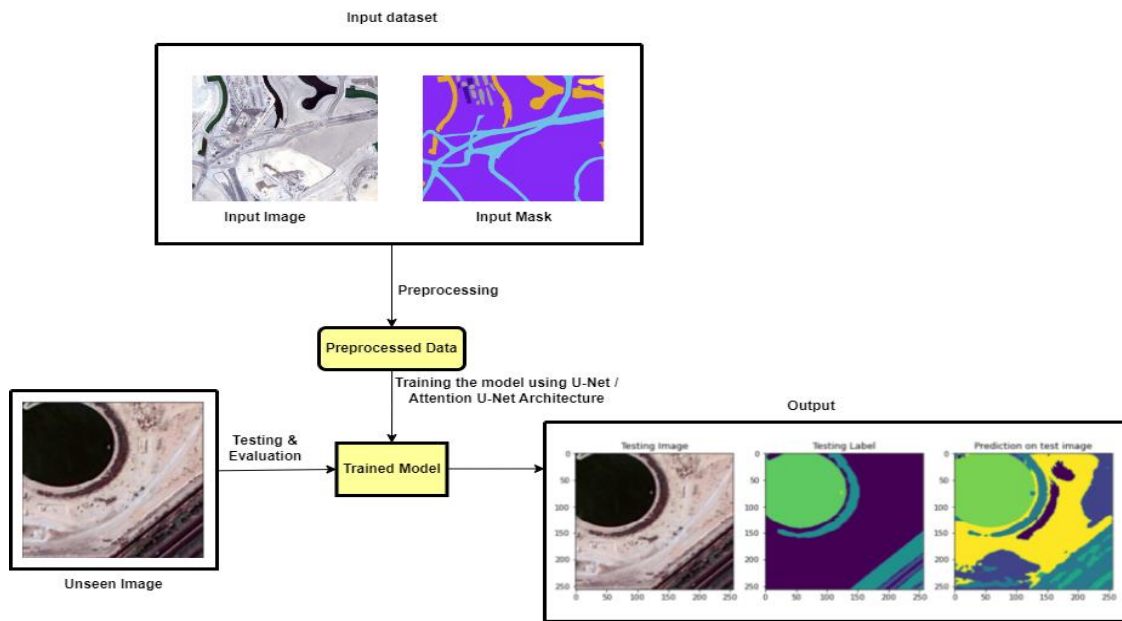


Fig. 2 : Block representation of the intended model

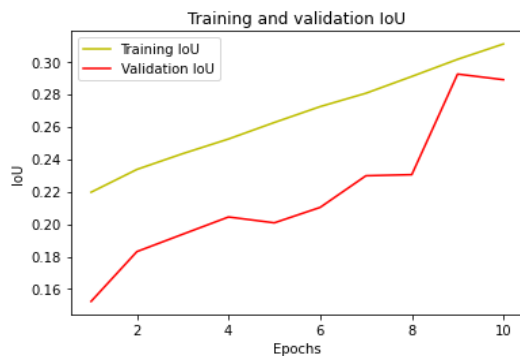


Fig. 3: Comparison of Training and Validation IoU graph of U-Net and Attention Res U-Net.

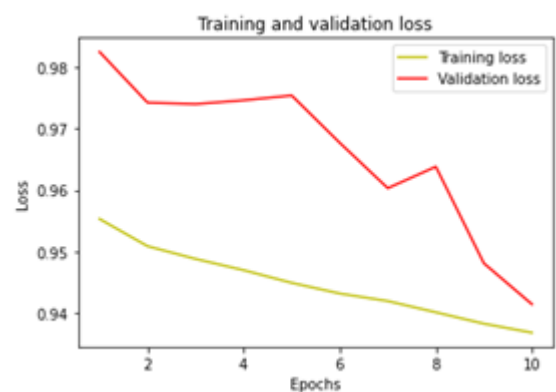


Fig. 4: Comparison of Training and validation loss graph of U-Net and Attention Res U-Net.

IV. RESULTS

The U-Net and the **Attention Res U-Net** and Attention U-Net are three neural network structures which are assessed in this comparison. Our goal is to determine which architecture excels, at segmenting data based on its semantic meaning.

Experimental Metrics and Comparison:

In order to examine the effectiveness of three designs the U-Net, **Attention Res U-Net** and Attention U-Net in semantic aerial data segmentation, we conducted experiments. Following ten epochs of U-Net model code execution, the accuracy percentage and training

loss are found to be 53.65 and 0.4755, respectively, in Attention U-Net framework and 0.4595 and 53.69, respectively, and **0.4216 and 53.72, respectively, in Attention Res U-Net**. Additionally, the program was run for 100 epochs, The training loss and the accuracy percentage in U-Net, **Attention Res U-Net**, and Attention U-Net is shown in the below table. Figure 5 represents the segmentation result of the U-Net framework, while Fig 6 illustrates the segmentation result of the Attention U-Net design. **The segmentation result of the Attention Res U-Net model is shown in Fig 7.**

Framework	Epochs	% of Accuracy	% of Mean IoU (mIoU)	Validation loss	Training loss
U-Net	10	53.65	39.84	1.905	0.4755
Attention U-Net	10	53.69	39.98	2.124	0.4595
Attention Res U-Net	10	53.72	42.97	1.786	0.4216
U-Net	100	83.62	61.28	0.6520	0.1905
Attention U-Net	100	85.99	61.32	0.4573	0.1136
Attention Res U-Net	100	86.42	63.85	0.4420	0.1058

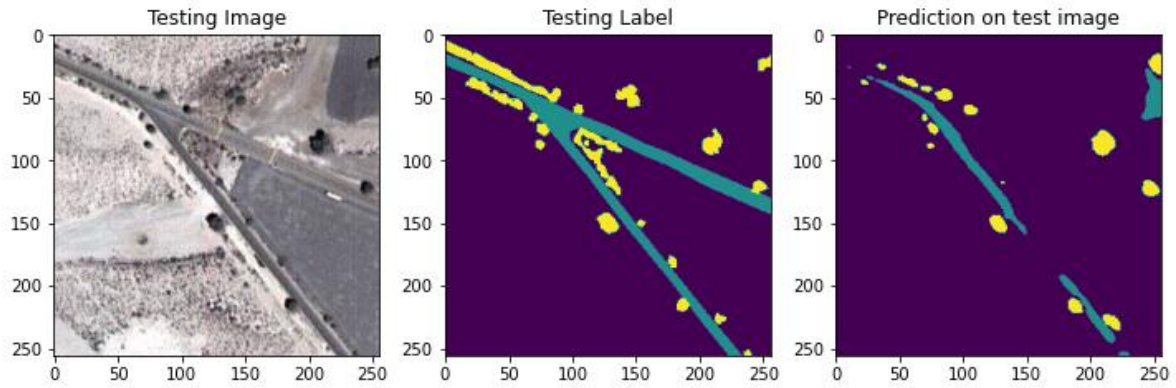


Fig . 5: Instance of Testing image(left),Testing label(middle) and Prediction on test image (right) of U-Net model.

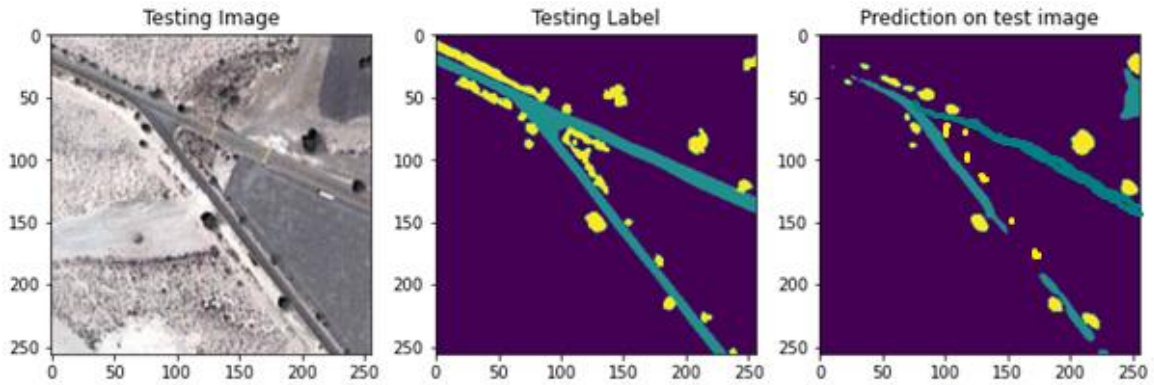


Fig. 6: Instance of Testing image(left),testing label(middle) and Prediction on test image (right) of Attention U-Net framework.

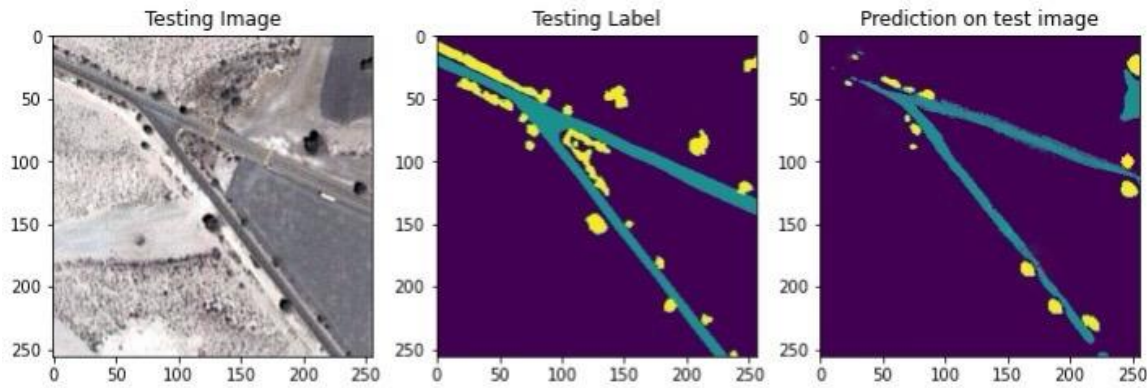


Figure 7: Instance of Testing image (left), Testing Label (middle) and Prediction on Test image (right) of Attention Res U-Net design.

V.CONCLUSION

Based on research findings we have discovered the feasibility **Attention Res U-Net** framework, in accurately dividing aerial data. By combining attention techniques with the U-Net architecture we were able to effectively handle and enhance spatial context preservation. Our comprehensive evaluation metrics, including accuracy, IoU scores and mIoU clearly showcased the models effectiveness. Beyond its applications our study has implications for environmental monitoring and urban planning. Looking ahead we anticipate advancements, in these approaches by integrating data and enhancing video based aerial analysis capabilities. Our ultimate objective is to advance machine learning's capabilities in analysing images for a conscious and environmentally friendly community.

VI.CONFLICT OF INTEREST

The contributors acknowledged no disagreements of concern.

VII.REFERENCES

- [1] Younis, Hameed Abdul-Kareem, and Marwa Kamel Hussien. "Multi-Frame Video Compression Scheme Using Three Step Search (TSS) Matching Algorithm." *The islamic college university journal* 29 (2014).
- [2] Younis, Hameed Abdul-Kareem, and Marwa Kamel Hussein. "Adaptive Video Compression Technique Based on Wavelet Transform and NTSS Matching Algorithm." *Journal of Education for Pure Science* 4.1 (2014).
- [3] Jetley, Saumya, et al. "Learn to pay attention." *arXiv preprint arXiv:1804.02391* (2018).
- [4] Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation." *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015*, Proceedings, Part III 18. Springer International Publishing, 2015.
- [5] Darapaneni, Narayana, et al. "Semantic Segmentation of Solar PV Panels and Wind Turbines in Satellite Images Using U-Net." *2020 IEEE 15th International Conference on Industrial and Information Systems (ICIIS)*. IEEE, 2020.
- [6] Heinrich, Mattias P., and Ozan Oktay. "BRIEFnet: Deep pancreas segmentation using binary sparse convolutions." *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Cham: Springer International Publishing, 2017.
- [7] Greff, Klaus, Rupesh K. Srivastava, and Jürgen Schmidhuber. "Highway and residual networks learn unrolled iterative estimation." *arXiv preprint arXiv:1612.07771* (2016).
- [8] Ali, Khawlah Hussein, and Tianjiang Wang. "Learning features for action recognition and identity with deep belief networks." *2014 International Conference on Audio, Language and Image Processing*. IEEE, 2014..
- [9] Anderson, Peter, et al. "Bottom-up and top-down attention for image captioning and visual question answering." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018.
- [10] Bahdanau, Dzmitry, Kyunghyun Cho, and Yoshua Bengio. "Neural machine translation by jointly learning to align and translate." *arXiv preprint arXiv:1409.0473* (2014).
- [11] Niu, Ruigang, et al. "Hybrid multiple attention network for semantic segmentation in aerial images." *IEEE Transactions on Geoscience and Remote Sensing* 60 (2021): 1-18.
- [12] Cai, Jinzheng, et al. "Improving deep pancreas segmentation in CT and MRI images via recurrent neural contextual learning and direct loss function." *arXiv preprint arXiv:1707.04912* (2017).
- [13] Cerrolaza, Juan J., Ronald M. Summers, and Marius George Linguraru. "Soft multi-organ shape models via generalized PCA: A general framework." *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2016: 19th International Conference, Athens, Greece, October 17-21, 2016*, Proceedings, Part III 19. Springer International Publishing, 2016..
- [14] Irwansyah, Edy, Yaya Heryadi, and Alexander Agung Santoso Gunawan. "Semantic image segmentation for building detection in urban area with aerial photograph image using U-Net models." *2020 IEEE Asia-Pacific Conference on Geoscience, Electronics and Remote Sensing Technology (AGERS)*. IEEE, 2020.

[15] Fang, Jie, QingBiao Zhou, and Shuxia Wang. "Segmentation technology of nucleus image based on U-net network." Scientific Programming 2021 (2021).

[16] Hussien, Marwa Kamel, and Hameed Abdul-Kareem Younis. "Wavelet-Based Video Compression System Using Diamond Search (DS) Matching Algorithm." journal of kerbala university 1.1 (2013).

[17] Hu, Jie, Li Shen, and Gang Sun. "Squeeze-and-excitation networks." Proceedings of the IEEE conference on computer vision and pattern recognition. 2018.

[18] Milletari, Fausto, Nassir Navab, and Seyed-Ahmad Ahmadi. "V-net: Fully convolutional neural networks for volumetric medical image segmentation." 2016 fourth international conference on 3D vision (3DV). Ieee, 2016.

[19] Kamnitsas, Konstantinos, et al. "Ensembles of multiple models and architectures for robust brain tumour segmentation." Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: Third International Workshop, BrainLes 2017, Held in Conjunction with MICCAI 2017, Quebec City, QC, Canada, September 14, 2017, Revised Selected Papers 3. Springer International Publishing, 2018.

[20] Heinrich, Mattias P., Max Blendowski, and Ozan Oktay. "TernaryNet: faster deep model inference without GPUs for medical 3D segmentation using sparse and binary convolutions." International journal of computer assisted radiology and surgery 13 (2018): 1311-1320.